

Rahul Manoj, PhD

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Curious, growth-oriented, hands-on electronics engineer/researcher with 5+ years experience in technology development for medical and industrial applications. Passionate about translating research into impactful products.

KEY QUALIFICATIONS

- **Electronics Product Engineering:** System-level thinking, Experience across ideation, modelling, prototyping, testing, verification, validation, and technical documentation for complex instrumentation systems.
- **Sensors & Measurement Systems:** Hands-on experience in electronic system design and integration of sensors & transducers for complex sensing environments. Skilled in sensor characterization, data acquisition.
- **Algorithm Development, Data Analytics & Signal Processing:** Implemented signal processing pipelines, software prototypes and tools in Python, MATLAB, LabVIEW; uncertainty quantification, sensitivity analysis of models.
- **Prototyping & Laboratory Experiments:** Building functional prototypes, experiment setups, and test rigs. Skilled in troubleshooting, test & measurement techniques, hardware-software integration of electro-mechanical systems.
- **Leadership & Collaboration:** Initiated and led multi-institution technology collaborations by identifying opportunities, aligning stakeholder interests, and translating them into deliverable projects. Managed R&D teams of 20+ members.
- **Scientific Communication & IP:** Published high-impact journal articles, presented at international conferences, joint patent applications, joint grant applications, proposals for ethical approval of pre-/clinical studies.

PROFESSIONAL EXPERIENCE

Research Scientist (Post-doc) | Division of Biomechanics

2024 to Present

Dept. of Structural Engineering, Norwegian University of Science & Technology, Trondheim, Norway

- Developed an end-to-end modelling pipeline integrating ultrasound data with a hybrid physics/data-driven models, delivering a validated clinical decision-support tool for early diagnosis of heart-failure pathologies.
- Formulated non-invasive calibration-free methods of blood pressure measurements combining cardiovascular biomechanics with advanced signal processing algorithms.
- Established regulatory-aligned V&V workflow as per ASME V&V 40, for system verification, model validation via detailed statistical analysis, uncertainty quantification and global sensitivity analysis using stochastic approaches.

Researcher | Advanced Cardiovascular Technologies Laboratory

2017 to 2024

Dept. of Electrical Engineering, Indian Institute of Technology (IIT) Madras, Chennai, India

- Engineered multi-sensor measurement systems integrating optical, force/pressure, and acoustics (ultrasound); performed sensor characterization, drift analysis, environmental sensitivity testing, and calibration.
- Integrated analog/digital electronics for low-noise acquisition and digital signal processing from photodiodes, strain gauges (capacitive and resistive), and ultrasound transducers (A-Mode, B-Mode, Doppler Flow, Plane Wave Imaging).
- Developed algorithms enabling robust extraction of noisy real-world biomedical signals and improved the measurement accuracy and precision of clinically relevant matrices for cardiovascular disease diagnosis.
- Build laboratory setups and test rigs (semi-automated electromechanical systems) for V&V studies; to validate sensors, sensitivity analysis, and to evaluate signal conditioning techniques in complex measurement environments.

Healthcare Technology Innovation Centre, IIT Madras Research Park, Chennai, India

- Co-designed and built a functional prototype of a modular and automatic time-scheduled body weight measurement system for bedridden patients, that can be integrated with hospital beds (Collaborator: Stryker®, India).
- Developed software and integrated electronics, sensors and actuators for an in-house designed automated cover slipper for large-format (6" X 8") histopathology human brain section slides.
- Co-drafted the medical device software lifecycle document for an in-house developed cardiovascular health assessment device ARTSENS® Plus as per IEC 62304 (Collaborator: VASC RISK LLC, USA).

SKILLS & EXPERTISE

- Technology Development & Validation
- Programming: LabVIEW | MATLAB | Python
- Hardware Prototyping | Test & Measurement
- Modelling | Machine Learning
- Uncertainty Quantification | Sensitivity Analysis
- Embedded Programming | SPICE Simulations
- Electronics | PCB Design | Signal Processing
- Git | Linux | Windows | MS Office | AI Tools
- Critical Thinking | Problem Solving
- Collaboration | Project Management
- Scientific Communication | Patents

SELECTED ACHIEVEMENTS

- Authored six scientific articles in peer reviewed, high impact instrumentation, biomedical, signal processing journals and over 40 international conference proceedings on algorithm development and instrumentation.
- Filed 5 patents in total as a co-inventor in India, EU and US. Granted: 1, Published: 4
- Won the popular prize for my pitch deck presentation for the *International Modelling Challenge: Digital Twin Builder for Health Incubator* by iSi Health KU Leuven, Belgium (June, 2025).

EDUCATION

- **M.S. & Ph.D.** in Electrical Engineering
(2017-2024) (*Specialised in Electronic Systems Design & Instrumentation*)
Indian Institute of Technology Madras (IIT M)
Chennai, Tamil Nadu, India
- **B. Tech.** in Electrical and Electronics Engineering
(2012-2016)
National Institute of Technology Calicut (NIT C)
Kozhikode, Kerala, India

PERSONAL PROFILE

- Community Outreach: Member of technical bodies, organising workshops, reviewer for scientific journals.
- Hobbies & Interests: Swimming, outdoor activities & travelling, science blogging, electronics hobbyist.
- Languages: English (C2), Norwegian (A2/B1), Indian Languages: Malayalam (native), Hindi (C1), Tamil (A2).

REFERENCES

Prof. Leif Rune Hellevik
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Dr. Jayaraj Joseph
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